# Computer Devices

## Peripheral

A **peripheral** is a device that is used to put information into or get information out of the computer.

There are three different types of peripherals:

- Input, used to interact with, or send data to the computer

  These are: mouse, keyboards, scanners, barcode readers.

- Output, which provides output to the user from the computer, and these are monitors, printers, speakers
- Storage, which stores data processed by the computer such as external hard drives, flash drives, etc.

**Note**: Devices that exist outside the computer case are called external peripherals, or auxiliary components,

Devices that are inside the case such as internal hard drives or CD-ROM drives are also peripherals in technical terms and are called internal peripherals but may not be recognized as peripherals by laypeople.

## Common Peripherals

1. Input: Keyboard, Computer mouse, Graphic tablet, Touchscreen, Barcode reader, Image scanner, Microphone, Webcam, Game controller, Light pen, Scanner, Digital camera
2. Output: Computer display, Printer, Projector, Speaker
3. Storage devices: Floppy disk drive, Flash drive, Disk drive, Smartphone or Tablet, computer storage interface, CD/DVD drive
4. Input/Output: Modem, Network interface controller (NIC)

## Input Devices

In computing, an **input device** is a peripheral (piece of computer hardware equipment) used to provide data and control signals to an information processing system such as a computer or other information appliance. Examples of input devices include keyboards, mice, scanners, digital cameras and joysticks.

## Keyboards

A **keyboard** is a human interface device which is represented as a layout of buttons. Each button, or key, can be used to either input a linguistic character to a computer, or to call upon a particular function of the computer. They act as the main text entry interface for most users. Traditional keyboards use spring-based buttons, though newer variations employ virtual keys, or even projected keyboards. It is typewriter like device composed of a matrix of switches.

Examples of types of keyboards include:

- Keyer
- Keyboard
- Lighted Program Function Keyboard (LPFK)

## Pointing Devices



*Figure 1: Computer mouse*

Pointing devices are the most used input devices today. A **pointing device** is any human interface device that allows a user to input spatial data to a computer. In the case of mice and touchpads, this is usually achieved by detecting movement across a physical surface. Analog devices, such as 3D mice, joysticks, or pointing sticks, function by reporting their angle of deflection. Movements of the pointing device are echoed on the screen by movements of the pointer, creating a simple, intuitive way to navigate a computer's graphical user interface (GUI).

## Imaging and Input Devices



*Figure 2: Microsoft Kinect sensor*

Video input devices are used to digitise images or video from the outside world into the computer. The information can be stored in a multitude of formats depending on the user's requirement.

- Digital camera
- Digital camcorder
- Portable media player
- Webcam
- Microsoft Kinect Sensor
- Image scanner
- Fingerprint scanner
- Barcode reader
- 3D scanner
- Laser rangefinder
- Eye gaze tracker

Medical Imaging

- Computed tomography
- Magnetic resonance imaging
- Positron emission tomography
- Medical ultrasonography

**Audio Input Devices**

Audio input devices are used to capture sound. In some cases, an audio output device can be used as an input device, to capture produced sound.

- Microphones
- MIDI keyboard or another digital musical instrument

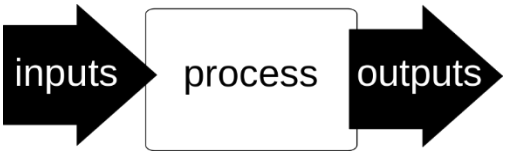
### Output Devices

An **output device** is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system (such as a computer) which converts the electronically generated information into human-readable form.

### Display Devices

A display device is an output device that visually conveys text, graphics, and video information. Information shown on a display device is called soft copybecause the information exists electronically and is displayed for a temporary period of time. Display devices include CRT monitors, LCD monitors and displays, gas plasma monitors, and televisions.

### Input/Output




Inputs are the signals or data received by the system, and outputs are the signals or data sent from it.

There are many input and output devices such as multifunction printers and computer-based navigation systems that are used for specialised or unique applications. In computing, input/output refers to the communication between aninformation processing system (such as a computer), and the outside world. Inputs are the signals or data received by the system, and outputs are the signals or data sent from it.

### Examples

These examples of output devices also include input/output devices. Printers and visual displays are the most common type of output device for interfacing to people, but voice is becoming increasingly available.

- Speakers
- Headphones
- Screen (Monitor)
- Printer

- Voice output communication aid
- Automotive navigation system
- Braille embosser
- Projector
- Plotter
- Television
- Radio

**Computer Memory**

In computing, **memory** refers to the devices used to store information for use in a computer. The term primary memory is used for storage systems which function at high-speed (i.e. RAM), as a distinction from secondary memory, which provides program and data storage that is slow to access but offer higher memory capacity. If needed, primary memory can be stored in secondary memory, through a memory management technique called "virtual memory." An archaic synonym for memory is store.

**Volatile Memory**



*Figure 3: DDR-SD-RAM, SD-RAM and two older forms of RAM*

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Volatile memory is computer memory that requires power to maintain the stored information. Most modern semiconductor volatile memory is either Static RAM (see SRAM) or dynamic RAM (see DRAM).

SRAM retains its contents as long as the power is connected and is easy to interface to but uses six transistors per bit. Dynamic RAM is more complicated to interface to and control and needs regular refresh cycles to prevent its contents being lost.

However, DRAM uses only one transistor and a capacitor per bit, allowing it to reach much higher densities and, with more bits on a memory chip, be much cheaper per bit.

SRAM is not worthwhile for desktop system memory, where DRAM dominates, but is used for their cache memories. SRAM is commonplace in small embedded systems, which might only need tens of kilobytes or less. Forthcoming volatile memory technologies that hope to replace or compete with SRAM and DRAM include Z-RAM, TTRAM, A-RAM and ETA RAM.

**Non-Volatile Memory**




*Figure 4: Solid-state drives are one of the latest forms of non-volatile memory*

Non-volatile memory is computer memory that can retain the stored information even when not powered. Examples of non-volatile memory include read-only memory (see ROM), flash memory, most types of magnetic computer storage devices (e.g. hard disks, floppy discs and magnetic tape), optical discs, and early computer storage methods such as paper tape and punched cards. Forthcoming non-volatile memory technologies include FeRAM, CBRAM,PRAM, SONOS, RRAM, Racetrack memory, NRAM and Millipede.